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Project Name : iPhone Sales Analysis

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Introduction Of Project

In today's fast-paced digital market, firms must understand consumer behavior and sales trends in order to stay competitive. With the popularity of mobile devices, notably iPhones, evaluating sales data on these platforms has become increasingly vital. This project intends to create a comprehensive sales analysis system based on iPhone data, utilizing the power of Python programming to find important insights and make instructed business decisions.

The goal of this project is to design and implement a data-driven solution that can process and analyze large datasets of iPhone sales, providing actionable recommendations to stakeholders. We built an effective and scalable system capable of handling complicated data analysis tasks by utilizing Python's data science ecosystem, which includes popular modules such as pandas, NumPy, and Matplotlib, Seaborn.

We worked with a dataset of iPhone sales, which will be cleaned, processed, and visualized to extract meaningful patterns and correlations.

Problem Statement

The purpose of this analysis is to better understand the sales success and consumer satisfaction of various iPhone models using important data such as sale prices, ratings, and customer reviews. By studying this data, we want to find trends and links between price, discounts, user ratings, and the amount of reviews, allowing us to make better business and marketing decisions.

Specifically, we aim to explore:

- How iPhone pricing affects the quantity of ratings and reviews.
- The relationship between user ratings and sales results.
- Does star ratings really reflect customer happiness and product popularity?

Algorithm used to solve the problem

Algorithm/Steps Used:

Data Collection & Preprocessing:

- Collected the dataset of various iPhone models with information such as sale price, number of ratings, number of reviews, star rating, and RAM.
- Preprocessing steps included handling missing values and ensuring data consistency (e.g., converting data types where needed).

Exploratory Data Analysis (EDA):

- Bar Chart (Visualization): Used to compare the sale prices of different iPhone models to understand pricing trends.
- Scatter Plot (Visualization): Used to investigate the relationship between the number of ratings and number of reviews, along with the influence of star ratings.
- Statistical Summary: Used basic statistics (mean, median, etc.) to summarize important aspects of the data, such as average ratings, price ranges, and distribution of customer reviews.

Correlations:

- Scatter Plots helped identify potential relationships between different variables (e.g., how the number of ratings relates to the number of reviews or star ratings).
- Visual inspection of data distributions and relationships through scatter plots enabled the identification of trends that would be crucial in making business decisions.

Technologies Used for iPhone Sales Data Analysis:

Python:

- Python was the primary programming language used for data analysis and visualization due to its powerful libraries for handling data, creating plots, and building presentations.

Pandas:

- A Python library used for data manipulation and analysis.
- It allows easy reading, cleaning, and manipulation of structured data like CSV files.

Matplotlib:

- A plotting library for creating static, animated, and interactive visualizations in Python.
- Used to generate various types of plots such as bar charts and scatter plots to visualize trends in iPhone sales data.

Seaborn:

- A Python data visualization library built on top of Matplotlib.
- It provides a high-level interface for creating attractive and informative statistical graphics.
- Used to create scatter plots and bar charts with added aesthetic appeal.

Jupyter Notebook:

- Although not explicitly mentioned, Jupyter Notebooks are often used to write, test, and run Python code interactively.

Screenshots of Project

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Set a theme for the plots
sns.set_theme(style="whitegrid")
```

In [2]:
```
# Load the dataset from a CSV file
data = pd.read_csv('C:/Users/jenne/OneDrive/Documents/OneDrive/Desktop/apple_products.csv')

# Show the first few rows of the dataset
data.head()
```

Out[2]:

	Product Name	Product URL	Brand	Sale Price	Mrp	Discount Percentage	Number Of Ratings	Number Of Reviews	Upc	Star Rating	Ram
0	APPLE iPhone 8 Plus (Gold, 64 GB)	https://www.flipkart.com/apple-iphone-8-plus-g...	Apple	49900	49900	0	3431	356	MOBEXRGV7EHHTGUH	4.6	2 GB
1	APPLE iPhone 8 Plus (Space Grey, 256 GB)	https://www.flipkart.com/apple-iphone-8-plus-s...	Apple	84900	84900	0	3431	356	MOBEXRGVAC6TJT4F	4.6	2 GB
2	APPLE iPhone 8 Plus (Silver, 256 GB)	https://www.flipkart.com/apple-iphone-8-plus-s...	Apple	84900	84900	0	3431	356	MOBEXRGVGETABXWZ	4.6	2 GB
3	APPLE iPhone 8 (Silver, 256 GB)	https://www.flipkart.com/apple-iphone-8-silver...	Apple	77000	77000	0	11202	794	MOBEXRGVMZWUHCBA	4.5	2 GB
4	APPLE iPhone 8 (Gold, 256 GB)	https://www.flipkart.com/apple-iphone-8-gold-2...	Apple	77000	77000	0	11202	794	MOBEXRGVPK7PFEJZ	4.5	2 GB

Screenshots of Project

```
In [5]: print(data.isnull().sum())
```

```
Product Name          0
Product URL           0
Brand                 0
Sale Price            0
Mrp                   0
Discount Percentage   0
Number Of Ratings     0
Number Of Reviews     0
Upc                   0
Star Rating           0
Ram                   0
dtype: int64
```

```
In [6]: print(data.describe())
```

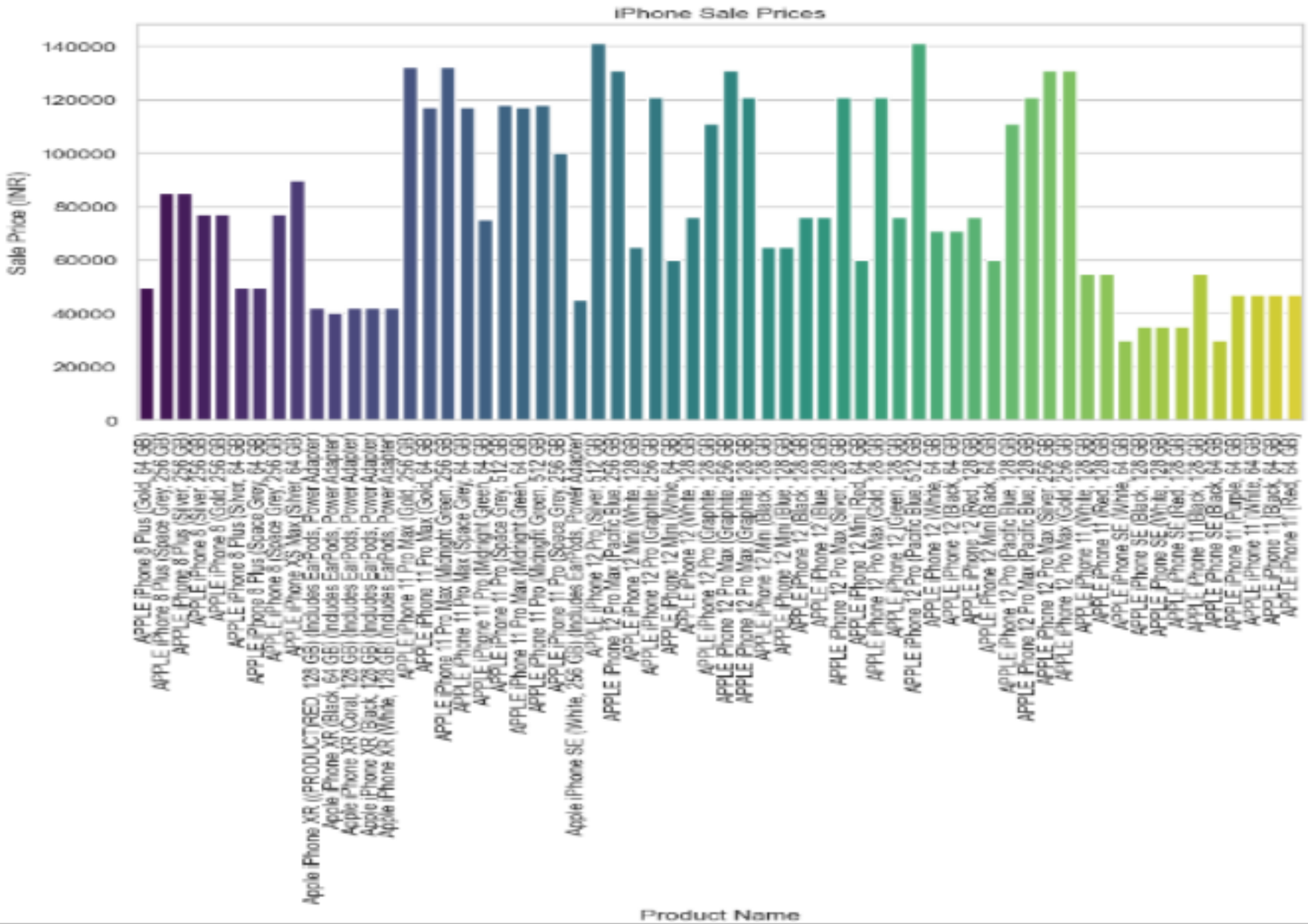
```
          Sale Price            Mrp  Discount Percentage  Number Of Ratings  \
count      62.000000      62.000000            62.000000          62.000000
mean    80073.887097   88058.064516             9.951613       22420.403226
std     34310.446132   34728.825597             7.608079       33768.589550
min     29999.000000   39900.000000             0.000000         542.000000
25%     49900.000000   54900.000000             6.000000         740.000000
50%     75900.000000   79900.000000            10.000000        2101.000000
75%    117100.000000  120950.000000            14.000000       43470.000000
max    140900.000000  149900.000000            29.000000       95909.000000

       Number Of Reviews  Star Rating
count          62.000000    62.000000
mean         1861.677419     4.575806
std          2855.883830     0.059190
min            42.000000     4.500000
25%            64.000000     4.500000
50%           180.000000     4.600000
75%          3331.000000     4.600000
max          8161.000000     4.700000
```

Screenshots of Project

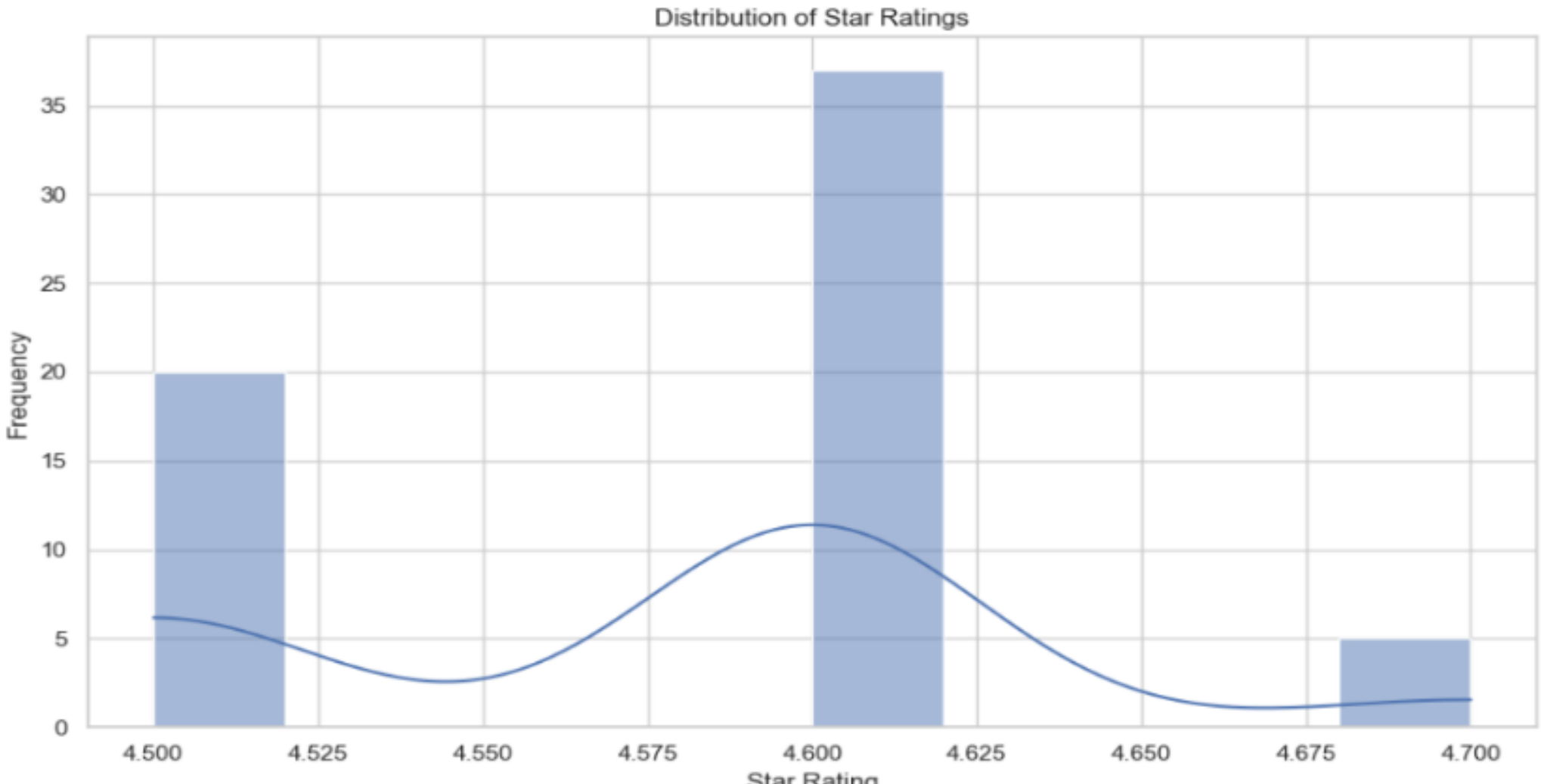
In [9]:
```
# Step 1: Price and Discount analysis
plt.figure(figsize=(10, 6))
sns.barplot(x='Product Name', y='Sale Price', data=data, palette='viridis')
plt.title('iPhone Sale Prices')
plt.xticks(rotation=90)
plt.ylabel('Sale Price (INR)')

plt.show()
```



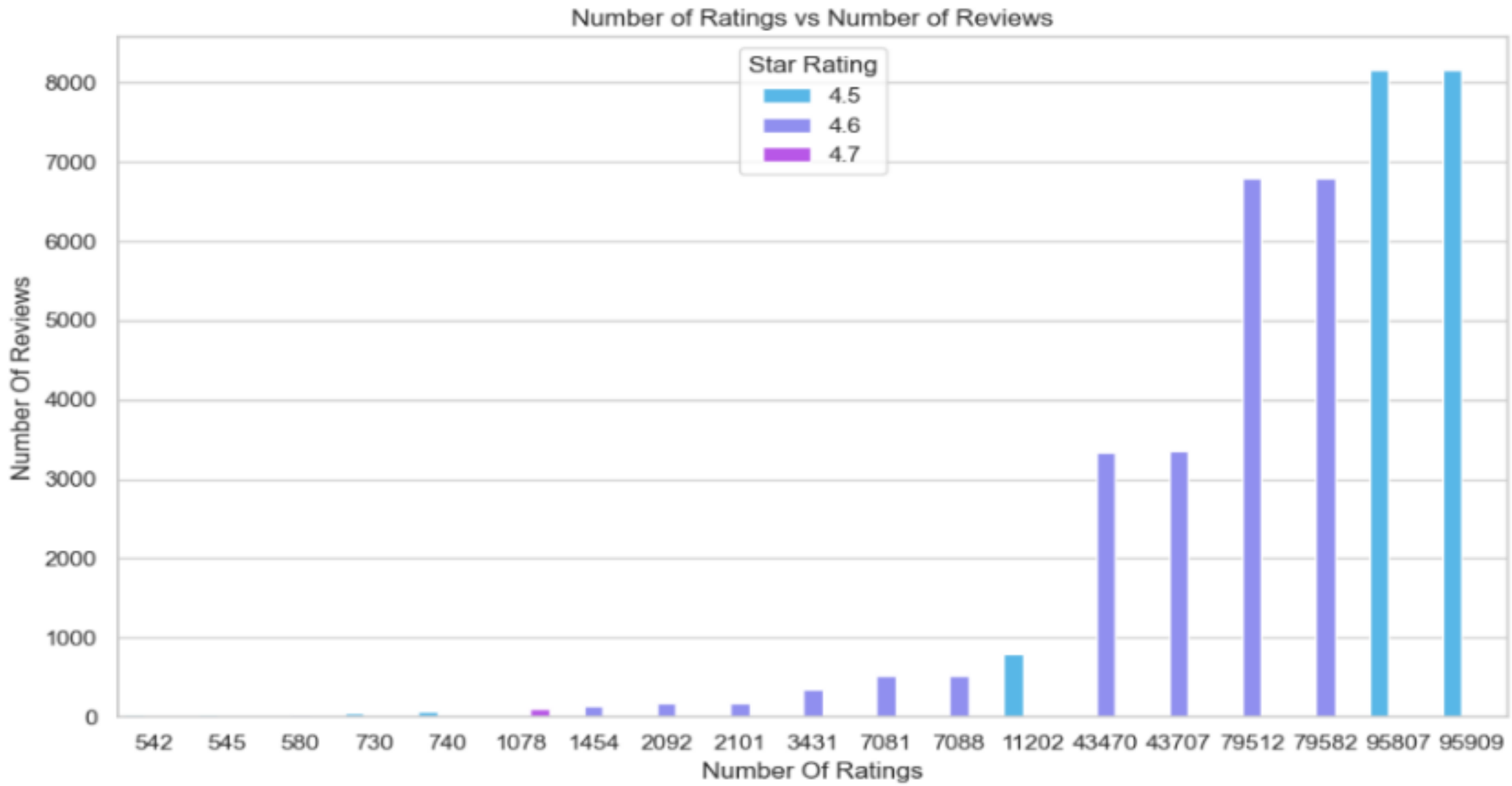
Screenshots of Project

In [10]:
```
plt.figure(figsize=(10, 6))
sns.histplot(data['Star Rating'], bins=10, kde=True)
plt.title('Distribution of Star Ratings')
plt.xlabel('Star Rating')
plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
```



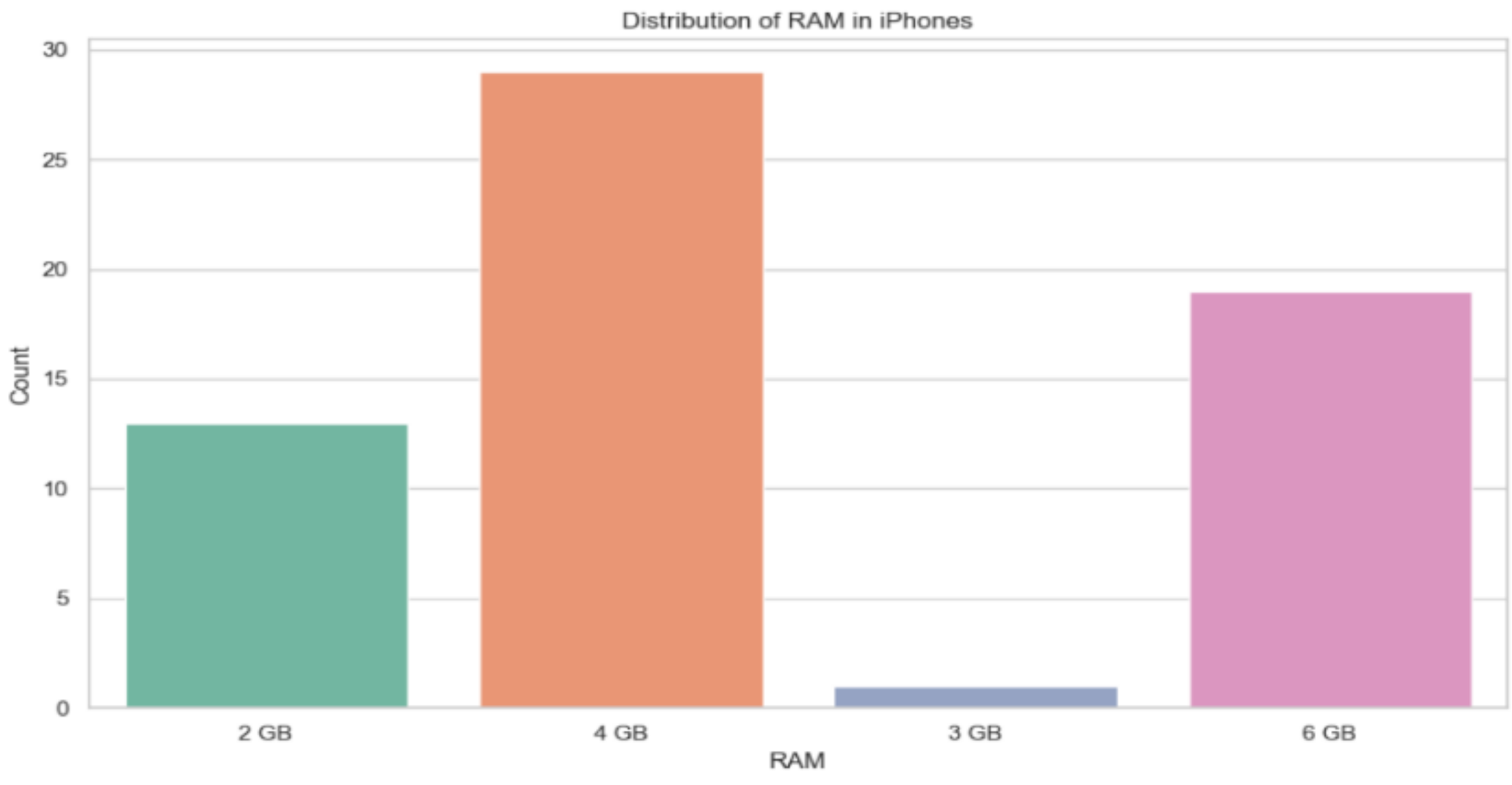
Screenshots of Project

```
In [14]: plt.figure(figsize=(10, 6))
         sns.barplot(x='Number Of Ratings', y='Number Of Reviews', hue='Star Rating', data=data, palette='cool')
         plt.title('Number of Ratings vs Number of Reviews')
         plt.xlabel('Number Of Ratings')
         plt.ylabel('Number Of Reviews')
         plt.tight_layout()
         plt.show()
```



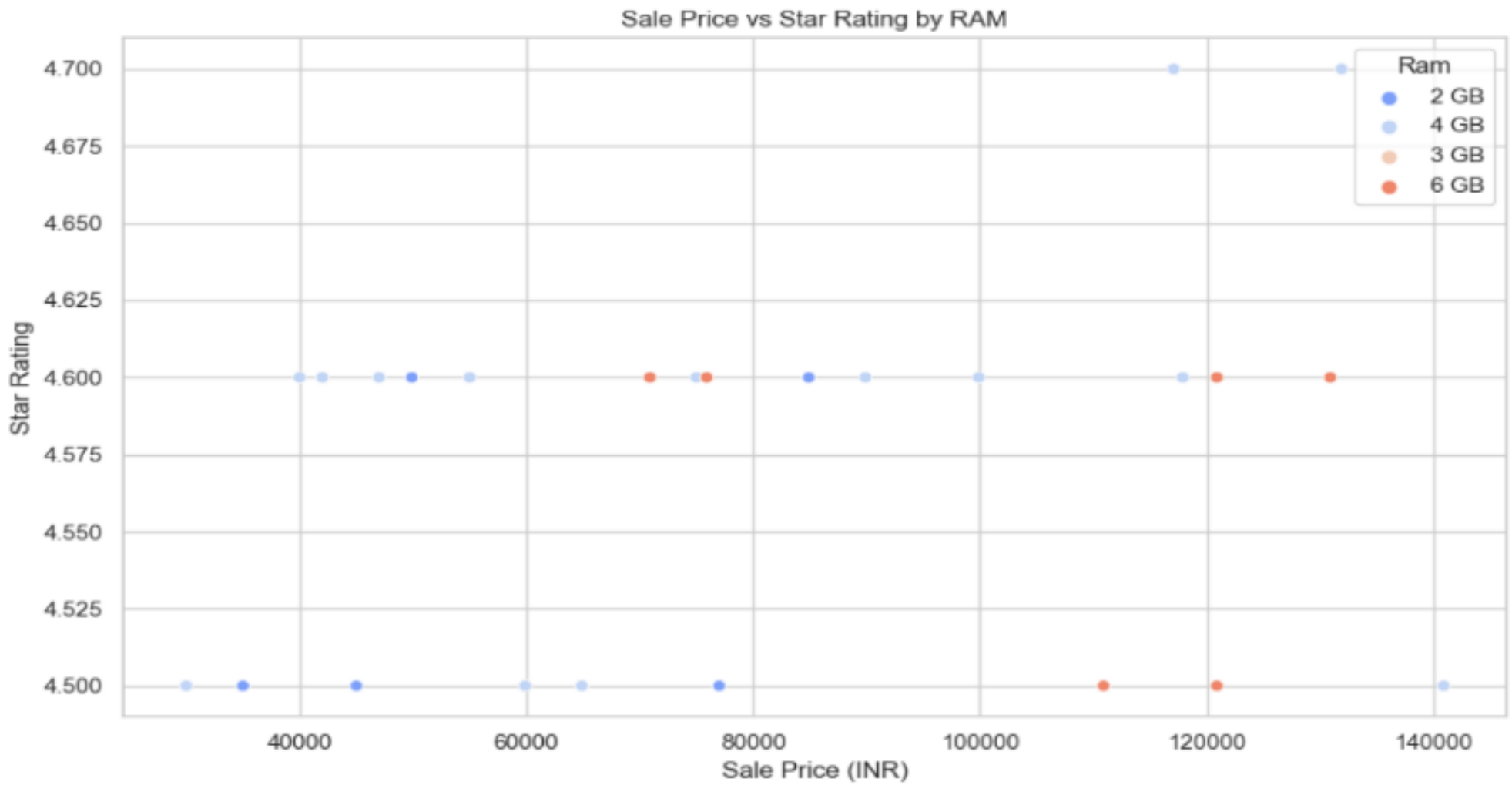
Screenshots of Project

In [12]:
```
plt.figure(figsize=(10, 6))
sns.countplot(x='Ram', data=data, palette='Set2')
plt.title('Distribution of RAM in iPhones')
plt.xlabel('RAM')
plt.ylabel('Count')
plt.tight_layout()
plt.show()
```



Screenshots of Project

In [13]:
```
# Step 5: Correlation between Star Rating and Sale Price
plt.figure(figsize=(10, 6))
sns.scatterplot(x='Sale Price', y='Star Rating', hue='Ram', data=data, palette='coolwarm')
plt.title('Sale Price vs Star Rating by RAM')
plt.xlabel('Sale Price (INR)')
plt.ylabel('Star Rating')
plt.tight_layout()
plt.show()
```



Future Scope

1.**Enhanced Data Collection**: Gather more detailed data on additional features like battery life and camera quality.

2.**Advanced Analysis Techniques**: Implement machine learning models to predict sales trends and customer ratings.

3.**Interactive Dashboards**: Develop interactive dashboards using tools like Plotly or Dash.

4.**Real-time Data Integration**: Integrate real-time data feeds to keep the analysis up-to-date.

5.**Comparative Analysis**: Compare Apple products with competitors to identify market strengths and weaknesses.

6.**Predictive Analytics**: Use predictive analytics to forecast future sales and market opportunities.

7.**User Experience Improvements**: Continuously improve the project based on user feedback.

Conclusion

In this project, we conducted a comprehensive analysis of Apple product data, focusing on various aspects such as sale prices, star ratings, and customer reviews. Through visualizations created using Matplotlib and Seaborn, we identified key trends and patterns, such as the distribution of RAM in iPhones and the correlation between sale prices and star ratings. This analysis provides valuable insights into customer preferences and product performance, which can be leveraged for strategic decision-making and future product development.

Additionally, the analysis highlighted the importance of detailed data collection and advanced analytical techniques in understanding market dynamics. By implementing machine learning models and interactive dashboards, future work can enhance the predictive power and user engagement of the project.

Overall, this project lays a solid foundation for ongoing analysis and improvement, offering a clear roadmap for future enhancements that can drive better business outcomes and customer satisfaction.

Reference

Books

Grolemund, G., & Wickham, H. (2017). *R for Data Science*. O'Reilly Media

VanderPlas, J. (2016). *Python Data Science Handbook*. O'Reilly Media.

Websites

DataCamp: Provides interactive courses on data science and analytics. DataCamp
Towards Data Science: A Medium publication with articles on data science, machine learning, and analytics. Towards Data Science
GitHub: Explore repositories like [iPhone Sales Analysis for practical projects and code examples]

Project on Data Analysis in iPhone Sales

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